
Chapter 7

Discussions and Conclusion

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Summary

EternaCare was developed to reduce delayed awareness when an older adult or independent individual living alone stops providing an expected safety signal. The solution combines a configurable Oren check-in, staged inactivity reminders, verified trusted contacts, manual SOS support, location-assisted alerts, virtual rewards, weather-responsive interaction, AI guidance and consent-controlled Legacy Planning. Flutter supports a consistent multi-platform client, while Supabase provides managed identity, relational data, security policies, private files and scheduled server execution.

Achievements

The project achieved its principal functional objectives. Check-ins now follow a one-to-168-hour rolling threshold instead of a fixed daily flag. A cloud worker evaluates every 15 minutes and persists the first, second and third missed-window stages. Contact phone verification, duplicate suppression, bounded SMS requests, stale-job recovery and server delivery state support safer escalation. Oren has hourly energy decay, user-scoped care state and serialized mutations that prevent lost rapid-tap updates. Dashboard refreshes reject stale overlapping responses and refresh on app resume. Rewards are fully virtual, with badge collection, voucher codes and a separate catalogue-admin route. Legacy Planning implements owner consent, funeral preferences, secret-aware notes, private documents, a 90-day trigger, a 24-hour owner-protection step and seven-day protected access. Supported sessions can be locked by asynchronously coordinated device biometrics, while the first-run guide and Settings module improve onboarding and accessibility.

Quality evidence includes 138 passing automated tests and 30.64 percent line coverage. The tests exercise important service, repository and widget paths, including threshold calculations, Oren decay and concurrency, OTP rules, Legacy timing, virtual rewards, guide/settings persistence and biometric prompt behaviour. This gives repeatable regression evidence, although it does not replace physical and provider acceptance testing.

Contributions

The main contribution is the integration of a low-friction virtual companion with an explicit, configurable safety heartbeat and a server-controlled escalation model. Separating care actions from the heartbeat reduces ambiguity, while virtual rewards support engagement without physical delivery administration. The Legacy design also contributes a consent-based staged release: the owner receives a final protection opportunity before time-limited access is offered to a verified primary contact.

Limitations and Future Improvements

EternaCare is not a medical device and cannot detect falls, unconsciousness, vital signs, death or the cause of inactivity. It does not automatically dispatch Malaysian emergency services. SMS and email delivery depend on provider accounts, carrier filtering and valid contact data. Android local scheduling is best-effort, GPS depends on permission and signal, and biometric authentication protects a restored local session rather than proving the identity of a trusted contact.

Future work should raise automated coverage, add provider sandbox and delivery-status integration tests, complete a structured physical-device matrix, add accessibility testing with older adults, introduce operational dashboards and alerting, support secure cross-device synchronization for Oren state, and perform professional privacy, legal and security reviews. Before production release, sensitive offline profile or medical caches should be removed when unnecessary or moved from unencrypted preferences to protected platform storage (OWASP Foundation, n.d.). The Android application identifier should also be changed from the development identifier, and direct SEND_SMS permission should be retained only after confirming Google Play eligibility and declaration requirements; otherwise, server-side SMS should remain the primary path (Google Play, n.d.). A production release should also define data retention, incident response, account recovery and support procedures.

Issues and Solutions

A major issue was that phone-only background execution could not guarantee threshold processing after the app was closed. This was addressed by moving authoritative evaluation to a 15-minute Supabase Cron worker while retaining WorkManager for local reminders. Duplicate SMS risk was reduced through persisted stages, idempotency keys and delivery outboxes; provider hangs and interrupted work were addressed with a 12-second timeout, retry states and five-minute stale-job recovery. Rapid Oren actions that could overwrite local state were serialized, and dashboard requests now discard stale overlapping results and refresh on app resume. OAuth users missing public profile rows were addressed with profile bootstrap logic. Private Legacy files were protected with RLS, private Storage and short-lived signed URLs. Repeated biometric prompts were addressed through asynchronous request coalescing and lifecycle-aware gating.

External delivery and platform behaviour remain the most important unresolved acceptance risks. The project therefore distinguishes a successful provider request from confirmed receipt and labels physical SMS, email, GPS, Android lifecycle and biometric-sensor tests separately from automated evidence.

Conclusion

The completed prototype demonstrates that a friendly check-in experience can be combined with explicit safety rules, server scheduling and privacy-conscious Legacy controls. Ethernacare provides useful support and earlier notification, but it must be presented honestly as a supplementary well-being system rather than a guaranteed emergency or medical service. Within those boundaries, the project satisfies its main academic objectives and provides a maintainable foundation for further validation and development.

8 References

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Project Title

Appendices

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Appendices

Appendix A: User Guide

Account and Setup

Install or launch Ethernacare, create an account or continue with Google, complete email confirmation where required, accept the Terms and Conditions, complete the profile, verify the user phone and add a verified primary trusted contact. The first completed setup opens an eight-step guide covering Home, History, Contacts, Rewards, AI Guidance, Legacy Planning, Profile and safety boundaries. The guide can be skipped and replayed later from Profile or Settings. No shared demonstration password is embedded in the application.

Daily Operation

Open Home and review Oren's status and the Safety Monitor. Tap Oren when the rolling threshold allows a check-in. Feed and Play change Oren care state but do not count as a safety heartbeat. Use History to inspect successful timestamps, Contacts to manage the verified safety network, Rewards to collect virtual items and Profile to change threshold, escalation and biometric settings. Open Profile > Settings to control local reminders, Oren sounds, text size and reduced motion, replay the feature guide or restore defaults.

Emergency and Legacy Functions

Use SOS only when a trusted contact should be notified; location is included only when permission and coordinates are available. Use the 999 action only for a real emergency because it opens an explicit call flow. In Legacy Planning, save funeral preferences, safe notes and supported private files, choose the authorised existing primary contact and enable release consent. Never store passwords, PINs, OTPs, recovery phrases or account credentials in Legacy Notes.

Appendix B: Developer Guide

Required Software

Flutter/Dart compatible with the project SDK constraint, Android Studio or Visual Studio Code, Android SDK for mobile builds, Visual Studio with Desktop development with C++ for Windows builds, Git, Node.js/npm for the Supabase CLI, and access to the configured Supabase project.

Backend Setup

Apply SQL migrations in supabase/migrations in filename order. Deploy all functions in supabase/functions, configure encrypted Twilio, email, AI and OTP secrets, verify authentication redirect URLs and providers, confirm private Storage policies, and inspect the 15-minute threshold and midnight Legacy Cron jobs. Confirm that production builds omit ENABLE_SAFETY_TESTS, set the final Android application identifier and include only approved platform permissions. Do not commit secret values or include them in screenshots.

Verification

Run flutter analyze, flutter test --coverage and the required release builds. The latest recorded automated run contains 138 passing tests with 2,515 of 8,208 executable lines covered (30.64 percent). Confirm database migration parity and Edge Function health. Complete physical Android checks for notification permission, background/resume behaviour, biometric success/cancel/retry, GPS capture and real SMS/email receipt. Record provider message IDs and errors without exposing tokens.